

A SIGNED-DIRECTION INVARIANT FOR TRIANGLE TILINGS, AND THE EXCLUSION OF PRIMES CONGRUENT TO 3 MODULO 4

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ABSTRACT. For a triangle T and a triangle ABC , an N -tiling is a dissection of ABC into N triangles each congruent to T . Erdős asked which N occur; a folklore conjecture is that no prime $N \equiv 3 \pmod{4}$ with $N > 3$ does (the value $N = 3$ occurs). By the classification of Laczkovich and the branch theorems of Beeson, the conjecture reduces to a tile with a $2\pi/3$ angle on a non-equilateral triangle, where a prime count forces ABC isosceles. For that isosceles case Beeson ruled out $N < 36$ and explicitly left open whether N can be prime (the smallest *known* tiling has 2673 tiles); we settle it. We introduce a translation-invariant, signed-direction tiling functional, linear in edge length and hence insensitive to non-edge-to-edge incidences, whose integrality forces, for an isosceles target, that $(c - a - b)/\sqrt{b}$ be an integer for a primitive triple with $c^2 = a^2 + ab + b^2$; an elementary lemma shows this never occurs. With a self-contained reduction of the scalene shapes, this completes the exclusion of primes $\equiv 3 \pmod{4}$ exceeding 3, conditional on the cited classification of the remaining branches. The arithmetic core is machine-checked in Lean 4 with Mathlib.

1. INTRODUCTION

Let T be a triangle (the *tile*). An N -tiling of a triangle ABC is a way of writing ABC as a union of N triangles, each congruent to T , with pairwise disjoint interiors. We do *not* assume the tiling is edge-to-edge: a vertex of one tile may lie in the relative interior of an edge of another. Erdős asked (reported by Soifer [12]) to determine the set of N for which some triangle admits an N -tiling by some tile. Every perfect square occurs (any triangle, by the n^2 subdivision), and Soifer observed that $n^2 + m^2$, $2n^2$, $3n^2$ and $6n^2$ occur as well; in particular $3 = 3 \cdot 1^2$ occurs. The outstanding question is whether the primes $\equiv 3 \pmod{4}$ that exceed 3, which are exactly the odd primes that are neither sums of two squares nor of the form $3n^2$, are excluded.

Theorem 1. *No prime $N \equiv 3 \pmod{4}$ with $N > 3$ is a number of congruent triangles into which some triangle can be cut. In particular, no triangle can be cut into 19 congruent triangles.*

The restriction $N > 3$ is necessary: an equilateral triangle joined to its centre splits into 3 congruent triangles, so $N = 3$ occurs.

The proof rests on Laczkovich’s classification of triangle tilings [9, 10], organized by Beeson across a series of papers [1, 2, 3, 4, 5]. For a prime number of tiles every branch of the classification is excluded by an existing theorem, with one exception: a tile having a $2\pi/3$ angle, tiling a non-equilateral triangle. We treat that branch.

This branch is the active frontier. Zhang [7] classifies the non-equilateral $2\pi/3$ shapes into six sporadic similarity types (Section 2 below), constructs families of tilings for each, and conjectures that the constructed N , all composite, are the only ones; the present results prove the no-prime half of that conjecture for the isosceles type, and reduce the scalene types directly. For the isosceles type Beeson [3] proved that no tiling exists with $N < 36$ and *explicitly left open whether N can be prime*; the smallest tiling presently known has 2673 tiles (Herdt, reported in [3]). The invariant below settles the question, excluding *every* prime. The adjacent Erdős Problem 633,

on tilings into a square number of triangles, was recently settled by Beeson, Laczkovich and Zhang [8].

Two reductions prepare it. First, by the rationality theorem of Beeson and Zhang [6], such a tile has commensurable sides, so after scaling we may take it integer-sided with $c^2 = a^2 + ab + b^2$. Second, an elementary area-and-incidence argument shows that a *prime* number of such tiles forces ABC to be *isosceles* (Section 3).

The heart of the paper is Section 4. The natural parity argument for a $2\pi/3$ tile (a black/white colouring of the tiles) fails, because a vertex of the tiling may be surrounded by an even number of tiles, so no consistent two-colouring exists. We replace it by a functional Φ that assigns to a directed edge of direction $\theta = j\frac{\pi}{3} + k\alpha$ the weight $(\text{length}) \cdot (-1)^j$. Two features make it work: $\Phi(\theta + \pi) = -\Phi(\theta)$, so interior edges cancel; and the weight is *linear in edge length*, so the cancellation is unaffected by non-edge-to-edge incidences, where one long edge meets several shorter ones summing to it. Summing over the tiles, Φ of the boundary equals an integer multiple of a fixed tile-value $c + a - b$. For an isosceles target this forces $(c - a - b)/\sqrt{b} \in \mathbb{Z}$, which never holds (Section 5). Assembling the branches proves Theorem 1 (Section 6).

The combinatorial claims are verified by the scripts in the accompanying repository, and the entire arithmetic layer, including Lemma 6 and the reduction of Section 3, is formalized and machine-checked in Lean 4 with Mathlib; see Section 7.

2. PRELIMINARIES

The classification. Fix a prime $N = p > 3$ with $p \equiv 3 \pmod{4}$. By Laczkovich [9, 10], every N -tiling of a triangle falls into one of finitely many types of pairs (ABC, T) . The relevant dichotomy, in Beeson's organization [4], is: either the angles of T are pairwise commensurable with π , or they are not. If they are commensurable, then N is a square, a sum of two squares, or 2, 3 or 6 times a square [4, Thm. 3 and Table 3]; but $p > 3$ with $p \equiv 3 \pmod{4}$ is a square only if p is one (it is not), a sum of two squares only if $p \not\equiv 3 \pmod{4}$, and $2n^2, 3n^2, 6n^2$ only if p is even or $3 \mid p$ (here p is odd and $p > 3$), so p is none of these (it is precisely for $p = 3 = 3 \cdot 1^2$ that this step would fail). If the angles are not commensurable, then, after the right-angle and similar-tile cases, which give N a sum of two squares or $N = n^2$ [1, 11], the tile has a special angle and falls into one of: $3\alpha + 2\beta = \pi$, for which N is not prime [2]; an isosceles tile with $\gamma = 2\alpha$, for which N is not squarefree [3, Thm. 11.7]; or the tile has an angle equal to $\pi/3$ or $2\pi/3$. The case of a $\pi/3$ or $2\pi/3$ tile on an *equilateral* triangle gives N not prime [5]. What remains for a prime N is a tile with a $\pi/3$ or $2\pi/3$ angle on a *non-equilateral* triangle. The vertex enumeration of Section 2 shows the $\pi/3$ case admits only equilateral or tile-similar ABC , already covered; so the sole remaining branch is a $2\pi/3$ tile, which we now fix.

The tile. Throughout, T has angles (α, β, γ) opposite sides (a, b, c) , with $\gamma = 2\pi/3$, hence $\alpha + \beta = \pi/3$; we treat the incommensurable case, α an irrational multiple of π . By Beeson–Zhang [6], T has commensurable sides, so we scale to

$$(a, b, c) \in \mathbb{Z}^3, \quad \gcd(a, b, c) = 1, \quad c^2 = a^2 + ab + b^2 \quad (1)$$

(the law of cosines at $\gamma = 2\pi/3$). We call such a triple a *primitive 120°-triple*. If a prime p divides two of a, b, c it divides the third (by (1)), so (1) gives $\gcd(a, b) = \gcd(a, c) = \gcd(b, c) = 1$. One computes

$$\sin \alpha = \frac{a\sqrt{3}}{2c}, \quad \cos \alpha = \frac{2b+a}{2c}, \quad \sin \beta = \frac{b\sqrt{3}}{2c}, \quad \cos \beta = \frac{2a+b}{2c}. \quad (2)$$

The direction grid. Set $G = \{j\frac{\pi}{3} + k\alpha : j, k \in \mathbb{Z}\} \pmod{2\pi}$. We claim every edge of every tile in a tiling points in a direction lying in G , for a single fixed reference. Fix the reference along an edge of one tile; its three edge-directions are then in G , since the tile angles $\alpha, \beta = \frac{\pi}{3} - \alpha, \gamma = \frac{2\pi}{3}$ all lie in $\mathbb{Z}\alpha + \mathbb{Z}\frac{\pi}{3}$. Now argue by adjacency. The tiling fills a connected region, so any tile

t' shares at least a boundary point P with the union of the tiles already placed; at P the surrounding corner angles each belong to $\{\alpha, \beta, \gamma\}$, and a directed edge of t' at P differs from a known grid direction at P by a sum of such angles, hence lies in G ; the remaining edges of t' then differ from it by $\mathbb{Z}$ -combinations of $\alpha, \frac{\pi}{3}$, so all of t' is in G . This uses only the angles meeting at a point and so applies verbatim at non-edge-to-edge incidences, where P lies in the relative interior of an edge (the angles there sum to π rather than 2π). Since α/π is irrational, $\frac{\pi}{3}$ and α are $\mathbb{Q}$ -linearly independent, so the pair (j, k) is determined by the direction (and $(-1)^j$ by the direction alone, as $2\pi = 6 \cdot \frac{\pi}{3}$).

Vertex types and shapes of ABC . A corner of ABC is filled by tile-corners: if $P, Q, R \geq 0$ (not all 0) of them are α -, β -, γ -corners, the corner measures $P\alpha + Q\beta + R\gamma = (P - Q)\alpha + (Q + 2R)\frac{\pi}{3}$. Write $(m, k) := (P - Q, Q + 2R)$. Since $Q \equiv k \pmod{2}$, $0 \leq Q \leq k$, and $P = m + Q \geq 0$, the pair (m, k) is *realizable* exactly when

$$k \geq 1 \text{ and } m \geq -k, \quad \text{or} \quad k = 0 \text{ and } m \geq 1. \quad (3)$$

The three corners of ABC have angles $m_i\alpha + k_i\frac{\pi}{3}$ summing to $\pi = 3 \cdot \frac{\pi}{3}$; as α is irrational this forces

$$\sum_i m_i = 0, \quad \sum_i k_i = 3. \quad (4)$$

By (3) each $m_i + k_i \geq 0$, and by (4) $\sum_i (m_i + k_i) = 3$; hence each $m_i + k_i \in \{0, 1, 2, 3\}$ and $m_i \in [-k_i, 3 - k_i] \subseteq [-3, 3]$. The enumeration is therefore finite and *independent of α* . Checking the finitely many triples $\{(m_i, k_i)\}$ satisfying (3), (4) and $0 < m_i\alpha + k_i\frac{\pi}{3} < \pi$ (done in exact arithmetic in Section 7) leaves exactly eleven shapes of ABC : the equilateral triangle; two isosceles triangles, with base angles α (apex $\alpha + 3\beta$) and β (apex $\beta + 3\alpha$); and eight scalene triangles, namely

$$F_1 = (\alpha, \alpha + \beta, \alpha + 2\beta), \quad F_2 = (2\alpha, 2\beta, \alpha + \beta), \quad F_3 = (\alpha, 2\alpha, 3\beta), \quad F_4 = (\alpha, 2\beta, 2\alpha + \beta),$$

their reflections under $\alpha \leftrightarrow \beta$ (with F_2 self-paired), and the tile-similar triangle $(\alpha, \beta, 2\pi/3)$. Since α/π is irrational, distinct pairs (m, k) give distinct angles, so the labels *equilateral*, *isosceles*, *scalene* attached to these types are unambiguous: no accidental coincidence of angles can occur, as any such coincidence would force $\alpha \in \mathbb{Q}\pi$. These types coincide with the six sporadic non-equilateral $2\pi/3$ similarity types of Zhang [7].

The area identity. If ABC is similar to a fixed shape whose primitive integer side-proportion vector is D (gcd of its entries 1), and ABC is N -tiled, then each side of ABC is a disjoint union of whole tile-edges (a tile meeting a straight portion of the boundary has a whole side on it), hence an integer; so the sides of ABC are $k \cdot D$ for an integer $k \geq 1$, and

$$N = \frac{\text{Area}(ABC)}{\text{Area}(T)} = k^2 N_0, \quad N_0 := \frac{\text{Area}(D)}{\text{Area}(T)}, \quad \text{Area}(T) = \frac{\sqrt{3}}{4} ab. \quad (5)$$

This is a necessary condition on N .

3. REDUCTION TO THE ISOSCELES CASE

Proposition 2. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. If T N -tiles a triangle and N is prime, then ABC is isosceles and not equilateral.*

Proof. By Section 2, the corner-type triple of ABC is the equilateral one, the tile-similar one, one of $F_1, \dots, F_4$ or their reflections, or one of the two isosceles ones. The equilateral case gives N not prime [5, Thm. 6]; the tile-similar case gives $N = n^2$ [1, 11], not prime. For the remaining non-isosceles shapes we compute N_0 from (5); in each case $N = k^2 N_0$ is composite.

F_2 and F_4 . The side-proportion vector of F_2 is $(a(a + 2b), b(2a + b), c^2)$ with the $\pi/3$ vertex between the first two sides, giving $N_0 = (a + 2b)(2a + b)$. For F_4 one checks the side-vector

$(ac, b(2a+b), (a+b)c)$ has gcd 1 (any common prime g divides c and then $g^2 \mid 3b^2$, forcing $g = 1$), giving $N_0 = (a+b)(2a+b)$. Both are products of two integer factors each at least 3, hence composite, so $N = k^2 N_0$ is composite.

F_3 . The law of sines gives the side-proportion vector $(ac^2, a(a+2b)c, 3ab(a+b))$, whose entries share the factor a ; dividing it out leaves $(c^2, (a+2b)c, 3b(a+b))$, and $N_0 = 3(a+b)(a+2b)$. This is divisible by 3, hence composite, so $N \equiv 0 \pmod{3}$ is composite.

F_1 . The side-vector is $(a, c, a+b)$ with the $\pi/3$ vertex between the sides a and $a+b$, so $N_0 = (a+b)/b$ and $N = k^2(a+b)/b$. As $\gcd(b, a+b) = 1$, integrality forces $b \mid k^2$, so $N = t(a+b)$ with $t = k^2/b \in \mathbb{Z}_{\geq 1}$. It remains to show $a+b$ is composite. Suppose $a+b = p$ is prime. Since $(a+b)^2 - c^2 = ab$ by (1), we have $ab = p^2 - c^2$, so a, b are the roots of $x^2 - px + (p^2 - c^2)$, whose discriminant $4c^2 - 3p^2$ must be a perfect square d^2 ; then $3p^2 = (2c-d)(2c+d)$. With $2c-d < 2c+d$ the factorizations of $3p^2$ into two positive factors are $(1, 3p^2)$, $(3, p^2)$, $(p, 3p)$, and in each the smaller root $\frac{p-d}{2}$ is not a positive integer: in the first, $d = (3p^2 - 1)/2 > p$, so the root is negative; in the second, $d = (p^2 - 3)/2$ and the root $\frac{p-d}{2} = -\frac{(p-3)(p+1)}{4}$ is negative for $p > 3$; in the third, $d = p$ gives $c = p$ and $ab = p^2 - c^2 = 0$. (The case $p \leq 3$ is impossible since $a+b \geq 8$ for a primitive 120° -triple.) Hence $a+b \geq 8$ is composite, and $N = t(a+b)$ is composite.

The reflected shapes are handled by $\alpha \leftrightarrow \beta$ (which swaps $a \leftrightarrow b$ and leaves each N_0 above of the same form). So a prime N forces the corner-type triple of ABC to be one of the two isosceles types; those have angles $(\varphi, \varphi, \pi - 2\varphi)$ with $\varphi \in \{\alpha, \beta\}$, and they are never equilateral ($\varphi = \frac{\pi}{3}$ would force $\beta = 0$ or $\alpha = 0$). Hence ABC is isosceles and not equilateral. $\square$

4. THE SIGNED-DIRECTION INVARIANT

Fix the reference so the direction grid is $G = \{j\frac{\pi}{3} + k\alpha\}$. Because $\beta = \frac{\pi}{3} - \alpha$, every grid direction also has the form $(j+k)\frac{\pi}{3} - k\beta$, so we may read off the coefficient of $\frac{\pi}{3}$ in two coordinate systems. Define, for a direction $\theta \in G$,

$$f_\alpha(\theta) = (-1)^j \quad (\theta = j\frac{\pi}{3} + k\alpha), \quad f_\beta(\theta) = (-1)^{j'} \quad (\theta = j'\frac{\pi}{3} + k'\beta).$$

Both are well defined (Section 2), and both satisfy

$$f(\theta + \pi) = -f(\theta), \tag{6}$$

since adding $\pi = 3 \cdot \frac{\pi}{3}$ increases the $\frac{\pi}{3}$ -coefficient by 3. They are genuinely different functions: $f_\beta = (-1)^k f_\alpha$.

For $f \in \{f_\alpha, f_\beta\}$ and a directed edge of length L and direction θ , assign the weight $L f(\theta)$. For a tile t with boundary traversed counterclockwise, put $C_f(t) = \sum_{\text{edges}} L f(\theta)$, and for the boundary of ABC put $\Phi_f(\partial ABC) = \sum_{\text{sides}} L f(\theta)$ (counterclockwise).

Lemma 3 (Cancellation). *For each $f \in \{f_\alpha, f_\beta\}$, $\sum_{\text{tiles}} C_f(t) = \Phi_f(\partial ABC)$.*

Proof. For a grid direction θ let $\Lambda(\theta)$ be the total length of all directed tile-edges of direction θ , summed over every tile (counterclockwise) and every edge; then, grouping the double sum $\sum_t C_f(t)$ by direction,

$$\sum_{\text{tiles}} C_f(t) = \sum_{\theta} \Lambda(\theta) f(\theta). \tag{7}$$

Split each tile-edge into its maximal sub-segments that are either interior to ABC or contained in ∂ABC ; this only refines the sums and changes nothing. Consider the finite set of maximal straight segments S carved out on the tiling's edges. If S lies in the interior of ABC , a neighbourhood of a generic point of S is a disc split by S into two half-discs, each tiled; the tiles on one

side meet S in directed edges of total length $|S|$ pointing one way, say θ , and those on the other side in directed edges of total length $|S|$ pointing $\theta + \pi$ (the two subdivisions may differ, a non-edge-to-edge incidence, but each side covers S exactly once because the tiles cover ABC with disjoint interiors). If instead $S \subseteq \partial ABC$, only the single tile inside ABC meets it, contributing length $|S|$ in the counterclockwise direction of ∂ABC . Writing $\Lambda = \Lambda_{\text{int}} + \Lambda_{\text{bd}}$ accordingly, interior segments give $\Lambda_{\text{int}}(\theta) = \Lambda_{\text{int}}(\theta + \pi)$ for every θ , while $\Lambda_{\text{bd}}(\theta)$ is the counterclockwise boundary length in direction θ . Substituting into (7) and pairing θ with $\theta + \pi$,

$$\sum_{\{\theta, \theta+\pi\}} \Lambda_{\text{int}}(\theta)(f(\theta) + f(\theta + \pi)) = 0$$

by (6), so $\sum_{\text{tiles}} C_f(t) = \sum_{\theta} \Lambda_{\text{bd}}(\theta)f(\theta) = \Phi_f(\partial ABC)$. $\square$

Lemma 4 (Tile value). *For every placement of T , $C_{f_\alpha}(t) = \pm(c+a-b)$ and $C_{f_\beta}(t) = \pm(c+b-a)$.*

Proof. Take a positively oriented placement and label the vertices V_1, V_2, V_3 in counterclockwise order with interior angles α, β, γ respectively; let e_i be the directed edge from V_i to V_{i+1} (indices mod 3). Then e_1, e_2, e_3 are opposite V_3, V_1, V_2 , so they have lengths c, a, b . Traversing counterclockwise turns left at V_{i+1} by the exterior angle $\pi - (\text{angle at } V_{i+1})$, so $\text{dir}(e_2) = \text{dir}(e_1) + (\pi - \beta)$ and $\text{dir}(e_3) = \text{dir}(e_2) + (\pi - \gamma)$. Now π adds 3 to the $\frac{\pi}{3}$ -coefficient j , while α adds 0, $\beta = \frac{\pi}{3} - \alpha$ adds 1, and $\gamma = \frac{2\pi}{3}$ adds 2; so $\pi - \beta$ adds $3 - 1 = 2$ and $\pi - \gamma$ adds $3 - 2 = 1$. Writing j_0 for the $\frac{\pi}{3}$ -coefficient of $\text{dir}(e_1)$, the three coefficients are $j_0, j_0 + 2, j_0 + 3$, so $f_\alpha = (-1)^{j_0}$ on e_1, e_2 and $-(-1)^{j_0}$ on e_3 . As e_1, e_2, e_3 carry lengths c, a, b ,

$$C_{f_\alpha}(t) = (-1)^{j_0}(c + a - b).$$

Any placement is obtained from this one by a grid rotation $m\alpha + n\frac{\pi}{3}$, which shifts every j by n and multiplies C_{f_α} by $(-1)^n$, possibly followed by a reflection, which reverses the traversal and negates C_{f_α} ; hence $C_{f_\alpha}(t) = \pm(c + a - b)$ in every case. Repeating in the β -frame (where $\gamma = \frac{2\pi}{3}$ adds 0 to j' and α adds 1) makes the a -edge the odd one out, giving $C_{f_\beta}(t) = \pm(c + b - a)$. $\square$

Corollary 5 (Integrality). *Each of $\frac{\Phi_{f_\alpha}(\partial ABC)}{c + a - b}$ and $\frac{\Phi_{f_\beta}(\partial ABC)}{c + b - a}$ is an integer. They are separate necessary conditions: each follows from Lemma 3 for its own f , so a single non-integer value forbids a tiling.*

Proof. By Lemmas 3 and 4, $\Phi_{f_\alpha}(\partial ABC) = \sum_t C_{f_\alpha}(t) = (c + a - b) \sum_t \varepsilon_t$ with $\varepsilon_t \in \{\pm 1\}$, and likewise for f_β . $\square$

5. THE ISOSCELES OBSTRUCTION

Lemma 6 (Non-integrality). *Let a, k be positive integers with $\gcd(a, k) = 1$, put $b = k^2$, and let $c > 0$ satisfy $c^2 = a^2 + ab + b^2$. Then $k \nmid (a + b - c)$.*

Proof. From $c^2 = a^2 + ab + b^2$ we get $c > a$ and, since $c^2 < (a + b)^2$, also $c < a + b$. Suppose $k \mid (a + b - c)$. As $b = k^2 \equiv 0 \pmod{k}$, this gives $c \equiv a \pmod{k}$; with $0 < c - a < b = k^2$ we may write $c = a + kt$, $1 \leq t \leq k - 1$. Substituting into $c^2 = a^2 + ak^2 + k^4$ and dividing by k ,

$$2at + kt^2 = ak + k^3, \quad \text{that is} \quad a(2t - k) = k(k - t)(k + t).$$

Then $k \mid a(2t - k)$, and $\gcd(a, k) = 1$ gives $k \mid (2t - k)$, hence $k \mid 2t$. But $2 \leq 2t \leq 2k - 2$ contains no multiple of k other than k itself, so $2t = k$. The left side is then $a(2t - k) = 0$, while the right side is $k \cdot \frac{k}{2} \cdot \frac{3k}{2} = \frac{3k^3}{4} > 0$, a contradiction. Hence $k \nmid (a + b - c)$. $\square$

Theorem 7. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. Then no prime number of copies of T tiles an isosceles, non-equilateral triangle.*

Remark 8. The irrationality hypothesis cannot be dropped: the commensurable tile $(\frac{\pi}{6}, \frac{\pi}{6}, \frac{2\pi}{3})$ tiles an equilateral (hence isosceles) triangle with $N = 3$, by the fan from the centre; that tile lives in the commensurable branch of the classification, where $N = 3n^2$ is allowed. The equilateral target for an *incommensurable* $2\pi/3$ tile is excluded for primes by Beeson [5], and is covered in the assembly of Section 6; it is not part of this theorem.

Proof. ABC has exactly two equal corner angles. Each corner angle is $m\alpha + k\frac{\pi}{3}$ (Section 2); since α/π is irrational, two corner angles are equal only if their pairs (m, k) coincide. So the corner-type triple of ABC has exactly one repeated type, and among the eleven triples of Section 2 exactly the two isosceles ones qualify. Hence the base angle is α or β . We use the invariant generated by the base angle, so that the equal sides land on $\frac{\pi}{3}$ -coefficient 3 (odd) and pick up $f = -1$.

If the base angle is α , then $ABC = k(c, c, 2b + a)$ (equal sides kc , base $k(2b + a)$): the base is at direction 0 ($f_\alpha = +1$) and the equal sides at $\pi - \alpha = 3\frac{\pi}{3} - \alpha$ ($f_\alpha = -1$), so $\Phi_{f_\alpha}(\partial ABC) = k(2b + a) - 2kc = k(2b + a - 2c)$. The tiling count is $N = k^2(a + 2b)/b$; since N is prime and $\gcd(a + 2b, b) = 1$, we have $k^2 = b$, i.e. $k = \sqrt{b}$. By Corollary 5 and $c^2 = a^2 + ab + b^2$,

$$\frac{k(2b + a - 2c)}{c + a - b} = \frac{c - a - b}{k} \in \mathbb{Z}, \quad k = \sqrt{b},$$

so $k \mid (a + b - c)$, contradicting Lemma 6. If the base angle is β , then $ABC = k(c, c, 2a + b)$, the equal sides are at $\pi - \beta$, and using f_β (for which the equal sides have $\frac{\pi}{3}$ -coefficient 3) the same computation gives, with $k = \sqrt{a}$, $(c - a - b)/k \in \mathbb{Z}$, again contradicting Lemma 6 (the integer $a + b - c$ is symmetric in a, b). In both cases no tiling exists. $\square$

Remark 9. The role of the two invariants is essential. For the base-angle- β shape, f_α places the equal sides at $\pi - \beta = \frac{2\pi}{3} + \alpha$, an *even* $\frac{\pi}{3}$ -coefficient, so f_α gives an integer value and does not obstruct; f_β does. Each invariant is a necessary condition in its own right (Corollary 5), so using the one that fails is legitimate, not a choice of convenient frame: on any genuine tiling both invariants return an integer.

6. RESOLUTION OF THE PROBLEM

Proof of Theorem 1. Let $p > 3$ be prime with $p \equiv 3 \pmod{4}$, and suppose some triangle is p -tiled. By the classification (Section 2) the tile is in one of the listed types, and every type other than a $2\pi/3$ tile on a non-equilateral triangle excludes such a p by an existing theorem [11, 1, 4, 2, 3, 5]. For a $2\pi/3$ tile, Proposition 2 forces ABC isosceles and not equilateral, and Theorem 7 excludes that. Hence no such tiling exists. Finally $19 > 3$ is prime and $\equiv 3 \pmod{4}$, so no triangle is cut into 19 congruent triangles. $\square$

Standing of the inputs. For the record, the external inputs to Theorem 1 and their publication status: Laczkovich's classification papers [9, 10] and the rep-tiling theorem of Snover–Waiveris–Williams [11] are refereed journal articles, and Beeson's isosceles paper [3] is accepted (to appear in *Integers*); Beeson's remaining branch papers [1, 2, 4, 5] and the Beeson–Zhang rationality theorem [6], which is load-bearing for the integrality of the tile, are arXiv preprints at the time of writing. Zhang's constructions [7] are cited for context only and carry no logical weight here. The parts original to this paper (Sections 3–5, together with the shape enumeration of Section 2) are self-contained given those inputs, with the arithmetic layer machine-checked (Section 7).

7. VERIFICATION

The finite and numerical claims are checked by the scripts in the repository (exact arithmetic). `verify_shapes.py` reproduces the eleven-shape enumeration of Section 2 and the values N_0 of

Section 3. `verify_invariant.py` confirms Lemma 4 ($C_f(t) = \pm(c + a - b)$ over all orientations of T), Lemma 3 (both sides agree on explicit subdivision tilings, and (6) cancels at a non-edge-to-edge incidence), and Lemma 6 (no counterexample among primitive 120° -triples up to $c \approx 9 \times 10^6$). As a positive control, both necessary conditions of Theorem 7 hold on the genuine 2673-tile isosceles tiling of Herdt [3]: for its tile $(5, 3, 7)$ and scale $k = 27$, $N = k^2(a+2b)/b = 2673$ and $M = k(2b + a - 2c)/(c + a - b) = -9$ are integers, as they must be on any true tiling.

The entire *arithmetic and combinatorial* layer of the proof is formalized and machine-checked in Lean 4 with Mathlib (`lean/Erdos634.lean`, thirteen theorems); every theorem depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`:

`shape_enumeration`: the eleven-triple completeness of Section 2 — realizable corner types with $\sum m_i = 0$, $\sum k_i = 3$ are exactly the eleven listed — as pure integer arithmetic; `k_not_dvd_sum_sub`, `M_not_int`: Lemma 6 ($((c - a - b)/\sqrt{b}) \notin \mathbb{Z}$); the full arithmetic of Theorem 7 in one master statement: no prime N satisfies the 120° -relation, the area equation $Nb = k^2(a + 2b)$, and the integrality $(c + a - b) \mid k(2b + a - 2c)$ simultaneously (`iso_reduction_identity`, `prime_count_forces_scale`, `no_prime_isosceles_count`); `add_not_prime`, `not_prime_of_two_le`, `F1_count_not_prime`–`F4_count_not_prime`: the scalene counts of Proposition 2 are composite ($a + b$ is never prime for a primitive 120° -triple, and the F_2, F_3, F_4 counts factor); `prime_three_mod_four_excluded`: a prime $p \equiv 3 \pmod{4}$ with $p > 3$ is neither a square, a sum of two squares, nor $2n^2$, $3n^2$ or $6n^2$ (the commensurable branch of Section 2), via Fermat’s two-squares theorem.

What is *not* formalized is exactly the geometry: the direction grid of Section 2 and Lemmas 3 and 4, which are proved in the text and checked numerically; Mathlib has no theory of planar dissections.

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